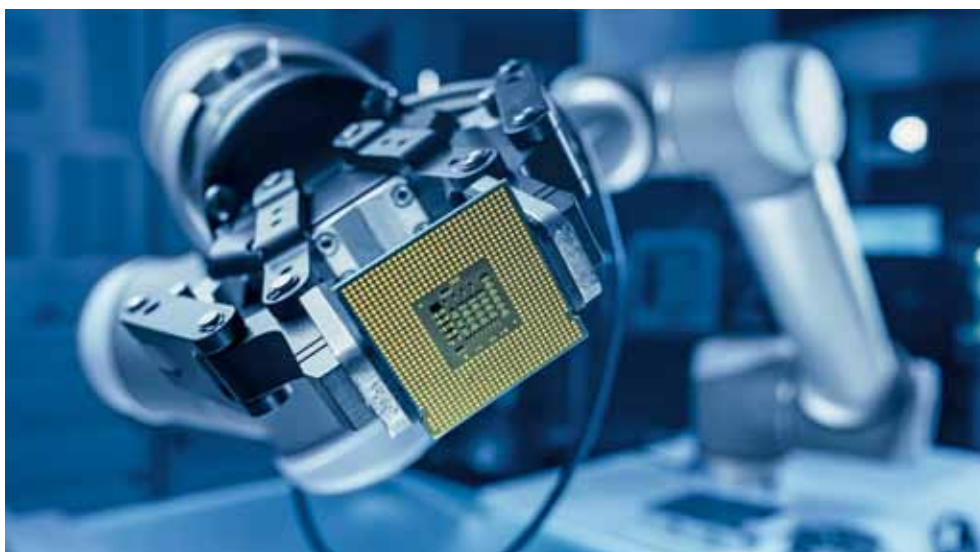


Application Of ARTIFICIAL INTELLIGENCE In The Manufacturing Sector

AI with ML swiftly predicts, minimises development time, and optimises synthesis, eliminating trial and error—for a future of advanced, efficient material manufacturing



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Artificial intelligence (AI) refers to the ability of a computer system to replicate human cognitive abilities, including learning and problem-solving skills. Although AI and machine learning (ML) are intricately related, they are distinct concepts. Machine learning is regarded as a subset of AI, indicating that it is a specialised area within the broader field of artificial intelligence. The relationship between artificial intelligence and machine learning provides significant advantages for businesses across diverse sectors, with constantly evolving opportunities and potential applications.

The integration of AI and ML technologies has revolutionised various industries, and material manufacturing

is no exception. This article explores the significant role AI and ML play in optimising and enhancing the entire material manufacturing process, from material design and synthesis to quality control and predictive maintenance. We delve into various applications, challenges, and future prospects, showcasing how these technologies are contributing to the evolution of the material manufacturing sector.

Material manufacturing has been a cornerstone of human civilisation, driving technological advancements and economic growth. Traditionally, material development and manufacturing processes relied on empirical knowledge, experimentation, and trial-and-error methods. However, with the advent of AI and ML, the